

An Analysis of Batch normalization

Batch normalization

$$\hat{X}_i = \frac{X_i - \mu_B}{\sqrt{\sigma_B^2 + \varepsilon}}$$

$$y_i = \gamma \hat{x}_i + \beta$$

Batch normalization -- Part 1

$$E[x(k)] = E_B[\mu_B(k)] \text{ and } \text{Var}[x(k)] = \frac{m}{m-1} E_B[(\sigma_B(k))^2].$$

Batch normalization -- Part 2

$$\|w_{\{t\}}\|_{\{S\}^{\{2\}}}\|\text{triangledown} f_{\{LH\}}(w_{\{t\}},g_{\{t\}})\|_{\{S^{\{-1\}}\}^{\{2\}}}\leq \{bigger\}1-\{\frac{\{\mu\}}{\{L\}}\}\{\bigger\}^{\{2t\}}\Phi^{\{2\}}\gamma_{\{t\}}^{\{2\}}(\rho_{\{w_0\}}-\rho^*)\}.$$

Combining these two inequalities, a bound could thus be obtained for the gradient with respect to $w \sim T_d$:

Batch normalization -- Part 3

ϵ is added in the denominator for numerical stability and is an arbitrarily small positive constant. The resulting normalized activation $x^{(k)}$ have zero mean and unit variance, if ϵ is not taken into account. To restore the representation power of the network, a transformation step then follows as

Batch normalization -- Part 4

$$\begin{aligned} & \|\mathbf{w} \sim (i) f(\mathbf{w} \sim t(i))\|_{S-1} \leq (1 - \mu_L) 2t C(\rho(\mathbf{w}_0) \\ & - \rho^*) + 2 - T s(i) \zeta \|\mathbf{b}_t(0) - \mathbf{a}_t(0)\|_{\mu_2} \end{aligned} \quad (2.15)$$

$$\|\textstyle\bigtriangleup_{\tilde{\mathbf{w}}^{(i)} f(\tilde{\mathbf{w}}^{(i)})} \|\mathbf{S}^{-1}\|^2 \leq \textstyle\bigg(1 - \frac{\mu_L}{\mu_U}\bigg)^{2t} C(\rho(\mathbf{w}_0) - \rho^*) + \frac{2^{-T s(i)} \zeta \|\mathbf{b}_t(0) - \mathbf{a}_t(0)\|_{\mu_2}}{\mu_U^2}.$$

Since the parameters of each hidden unit converge linearly, the whole optimization problem has a linear rate of convergence.

Batch normalization -- Part 5

$$y(k) = B N \gamma(k), \beta(k) \inf(x(k)) = \gamma(k) x(k) - E[x(k)] \sqrt{\text{Var}[x(k)] + \beta(k)}$$

$$y^{\{k\}} = B N_{\gamma^{\{k\}}} \frac{x^{\{k\}} - E[x^{\{k\}}]}{\sqrt{\text{Var}[x^{\{k\}}] + \epsilon}} + \beta^{\{k\}}, \text{ where } y(k) \text{ is passed on to next layers instead of } x(k).$$

Since the parameters are fixed in this transformation, the batch normalization procedure is essentially applying a linear transform to the activation function.

Batch normalization -- Part 6

$\lambda_1 - \rho(w_{t+1}) \rho(w_{t+1} - \lambda_2) \leq (1 - \lambda_1 - \lambda_2 \lambda_1 - \lambda_{\min})^2 \frac{\lambda_1 - \rho(w_t) \rho(w_t) - \lambda_2}{\rho(w_{t+1}) \rho(w_{t+1} - \lambda_2)} \leq \frac{\lambda_1 - \lambda_2}{\lambda_1 - \lambda_{\min}} \frac{\lambda_1 - \rho(w_t) \rho(w_t) - \lambda_2}{\rho(w_t) \rho(w_t - \lambda_2)}$, where λ_1 is the largest eigenvalue of B , λ_2 is the second largest eigenvalue of B , and $\lambda_{\min}$ is the smallest eigenvalue of B . In our case, $B = uu^T$ is a rank one matrix, and the convergence result can be simplified accordingly. Specifically, consider gradient descent steps of the form $w_{t+1} = w_t - \eta_t \nabla \rho(w_t)$ with step size $\eta_t = \frac{w_t^T S w_t}{2L \|\rho(w_t)\|}$, and starting from $\rho(w_0) \neq 0$, then

Batch normalization -- Part 7

$\hat{w}^{\wedge}(i) = c^{\wedge}(i) S - 1 u \{\displaystyle \hat{w}\}^{\wedge\{i\}} = \{\hat{c}\}^{\wedge\{i\}} S^{\wedge\{-1\}} u$, where $c^{\wedge}(i) \in R \{\displaystyle \hat{c}\}^{\wedge\{i\}} \in R$ is a scalar, $i = 1, \dots, m \{\displaystyle i=1,...,m\}$. This result could be proved by setting the gradient of $f_{NN} \{\displaystyle f_{NN}\}$ to zero and solving the system of equations. Apply the GDNP algorithm to this optimization problem by alternating optimization over the different hidden units. Specifically, for each hidden unit, run GDNP to find the optimal $W \{\displaystyle W\}$ and $\gamma \{\displaystyle \gamma\}$. With the same choice of stopping criterion and stepsize, it follows that

Batch normalization -- Part 8

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$$\begin{aligned} & \left(\left(\nabla_{\mathbf{y}_j} \mathbf{L} \right)^\top \nabla \mathbf{T} \nabla \partial \mathbf{y}_j \partial \mathbf{y}_j \left(\nabla_{\mathbf{y}_j} \mathbf{L} \right)^\top \right) \leq \gamma \mathbf{m} \sigma^2 \left(\nabla \mathbf{L} \nabla \partial \mathbf{y}_j \right)^\top \\ & \nabla \left(\nabla \mathbf{L} \nabla \partial \mathbf{y}_j \partial \mathbf{y}_j \right) \left(\nabla \mathbf{L} \nabla \partial \mathbf{y}_j \right) - \gamma \mathbf{m} \sigma^2 \left(\nabla_{\mathbf{y}_j} \mathbf{L}, \mathbf{y}_j \right)^\top \left| \left| \nabla \mathbf{L} \nabla \partial \mathbf{y}_j \right| \right|^2 \\ & \left\{ \left(\nabla_{\mathbf{y}_j} \mathbf{L} \right)^\top \nabla \left(\nabla \mathbf{L} \nabla \partial \mathbf{y}_j \partial \mathbf{y}_j \right) \left(\nabla_{\mathbf{y}_j} \mathbf{L} \right)^\top \right\} \leq \left(\frac{\gamma \mathbf{m} \sigma^2}{\left(\nabla_{\mathbf{y}_j} \mathbf{L} \right)^\top \nabla \left(\nabla \mathbf{L} \nabla \partial \mathbf{y}_j \partial \mathbf{y}_j \right) \left(\nabla_{\mathbf{y}_j} \mathbf{L} \right)^\top} \right) \left(\nabla_{\mathbf{y}_j} \mathbf{L}, \mathbf{y}_j \right)^\top \left| \left| \nabla \mathbf{L} \nabla \partial \mathbf{y}_j \right| \right|^2 \end{aligned}$$

indicates that the loss Hessian is resilient to the mini-batch variance, whereas the second term on the right hand side suggests that it becomes smoother when the Hessian and the inner product are non-negative. If the loss is locally convex, then the Hessian is positive semi-definite, while the inner product is positive if $\mathbf{g}_j^\top \hat{\mathbf{g}}_j$ is in the direction towards the minimum of the loss. It could thus be concluded from this inequality that the gradient generally becomes more predictive with the batch normalization layer. It then follows to translate the bounds related to the loss with respect to the normalized activation to a bound on the loss with respect to the network weights: